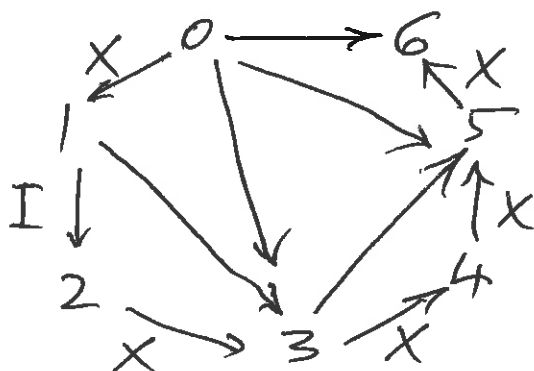


Our goal is to discover a nice model for the free skew-monoidal category $\mathcal{F}$ generated by a single object X . The objects are clearly words in the letters I and X , bracketed meaningfully pairwise. For example, there is the object

$$(((X(IX))(XX))X);$$

but we can write this as a triangulated polygon



where the edges $i \rightarrow i+1$ are labelled by I or X .

In other words, our objects are 2-cells

$$\begin{array}{ccc} \{0, n\} & & \\ 0 \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow T \\ \xrightarrow{\quad} \end{array} n & & \\ \{ (0,1), (1,2), \dots, (n-1, n) \} & & \end{array}$$

of the n -th oriental $\mathcal{O}_n$ together with a subset u of the ordinal

$$\underline{n} = \{0, 1, \dots, n-1\}$$

where $i \in u$ means the edge $(i, i+1): i \rightarrow i+1$

is labelled by I and the other edges by X . ^②

Using Huang-Tamari (1972), we can see that, for $S, T : \{0n\} \Rightarrow \{(01), \dots, (n-1, n)\}$, the following are equivalent:

(i) the bracketing corresponding to S is less than or equal to that corresponding to T in the Tamari order;

(ii) there exists a 3-cell $S \rightarrow T$ in $\mathcal{O}_n$;

(iii) $(i, j, k) \in S$ and $(i', j, k') \in T$ implies $i \leq i'$ and $k \leq k'$;

(iv) $(i, j, k) \in S$ and $(i', j, k') \in T$ implies $i \leq i'$;

(v) $(i, j, k) \in S$ and $(i', j, k') \in T$ implies $k \leq k'$.

But $b_n = \{0n\}$ and $e_n = \{(01), \dots, (n-1, n)\}$.

In the case $n=1$, we have the identity 2-cell

$$\begin{array}{ccc} & \{0,1\} = b_1 & \\ 0 & \xrightarrow{\quad} & 1 \\ & \Downarrow \alpha & \\ & \{01\} = e_1 & \end{array}$$

in $\mathcal{O}_1$ which may or may not have its

edge labelled by I ; that is, $u \subseteq \frac{1}{2}$ may be $u = \frac{1}{2}$ or $u = \emptyset$, respectively. ③

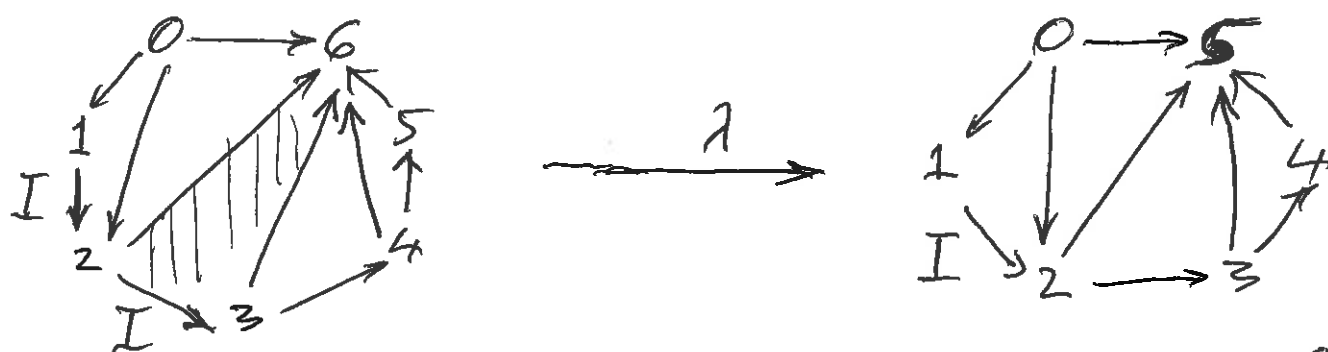
To clarify, we are going to take the objects of $\mathcal{F}$ to be pairs (u, T) where $u \subseteq \mathbb{N}$ and $T \in \mathcal{O}_n(0, n)(b_n, e_n)$ for $n > 0$.

In $\mathcal{F}$, there is a unique morphism

$$(u, S) \xrightarrow{\alpha} (u, T)$$

if and only if $S \leq T$ in the Tamari order. The morphisms of type λ and ρ are a little trickier.

Consider an example:



which we think of as degenerating the shaded triangle $(2\ 3\ 6)$ in the domain and renumbering appropriately. The reason this is of type λ is that the label I being lost is on the first morphism of the composable

edges in the triangle (236). In $\mathcal{F}$, we have ⁽⁴⁾

$$\begin{aligned} & (\{1, 2\}, \{(012), (026), (236), (346), (456)\}) \\ & \xrightarrow{\lambda} (\{1\}, \{(012), (025), (235), (345)\}) \end{aligned}$$

Clearly this involves the morphism

$$\sigma_2 : \underline{6} \longrightarrow \underline{5}$$

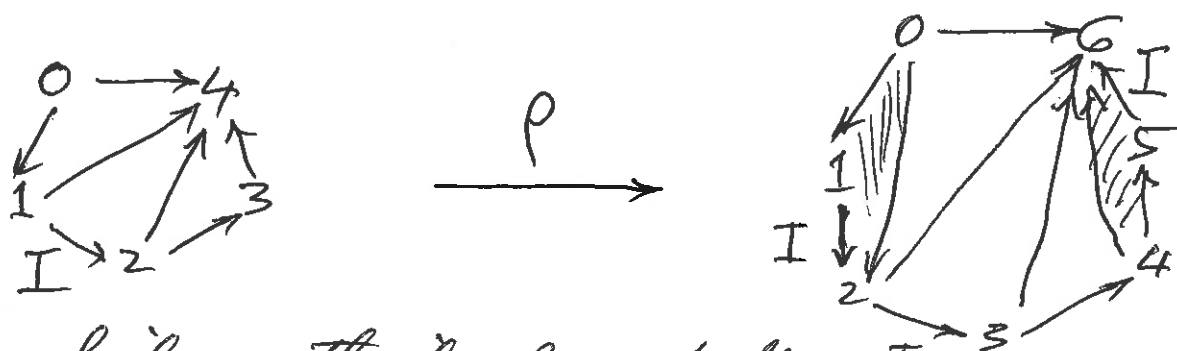
taking $0, 1, 2, 3, 4, 5$ to $0, 1, 2, 2, 3, 4$, respectively. So the morphism has the form

$$(u, S) \xrightarrow{\lambda} (v, T)$$

where $v = \sigma_2(u \setminus \{23\})$ and

$$T = \sigma_2(S \setminus \{(236)\}).$$

Consider another example:



which we think of as puffing up the edges (01) and (34) to triangles (012) and (456) . The reason this is of type ρ is that the label I is being put on the second morphism of the composable pair of a triangle. In $\mathcal{F}$, we

(5)

have

$$(\{1\}, \{(014), (124), (234)\}) \xrightarrow{\rho} (\{1, 2, 5\}, \{(012), (026), (236), (346), (456)\}).$$

Clearly this involves the morphism

$$\partial_{15} : \underline{4} \longrightarrow \underline{6},$$

taking $0, 1, 2, 3$ to $0, 2, 3, 4$, whose image does not contain 1 or 5. So the morphism has the form

$$(u, S) \xrightarrow{\rho} (v, T)$$

where $v = \partial_{15}(u) \cup \{1, 5\}$

$$T = \partial_{15}(S) \cup \{(012), (456)\}.$$

26 Nov 2012

Let us attempt a definition of general morphism of $\mathcal{F}$. For each $S: b_m \Rightarrow c_m$ in $\mathcal{O}_m$, define a function

$$t_S : \underline{m} \longrightarrow S$$

by

$$t_S(i) = (l_S(i), i, r_S(i)),$$

thereby also defining functions

$$l_S : \underline{m} \longrightarrow \underline{m} \quad \text{and} \quad r_S : \underline{m} \longrightarrow \{1, 2, \dots, m\}.$$

We say $i \in \underline{m}$ is of λ type when
 $l_S(i+1) = i$.

(6)

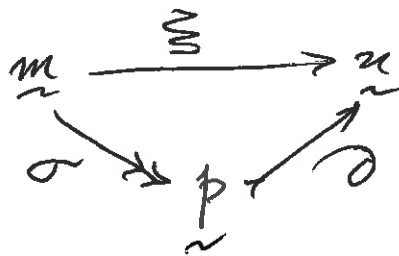
We say $i \in \underline{m}$ is of ρ type when
 $r_S(i) = i+1$.

For an order-preserving function

$$\xi : \underline{m} \longrightarrow \underline{n},$$

put $\xi_\lambda = \{i \in \underline{m} \mid \xi(i) = \xi(i+1)\}$ and
 $\xi_\rho = \{j \in \underline{n} \mid j \notin \text{im } \xi\}$.

Then



where

$$\begin{aligned}
 \sigma_\lambda &= \xi_\lambda & \text{and} & & \tau_\rho &= \xi_\rho \\
 \sigma_\rho &= \emptyset & & & \tau_\lambda &= \emptyset.
 \end{aligned}$$

A morphism $\xi : (u, S) \longrightarrow (v, T)$ in $\mathcal{F}$
 is an order-preserving function

$$\xi : \underline{m} \longrightarrow \underline{n}$$

(where $m = \#S$ and $n = \#T$) such that

there exists a 2-cell $L : b_m \Rightarrow c_m$ in $\mathcal{J}_m$ satisfying (17)

- $S \leq L$ in the Tamari order,
- $R = \xi(R \setminus \{t_R(i) \mid i \in u \setminus \xi_\lambda\}) \cup \{t_R(i) \mid i \in \xi_\rho\} \leq T$ in the Tamari order,
- $\xi_\lambda \subseteq u$, and ξ_λ consists of elements of λ type for L ,
- $\xi_\rho \subseteq v$, and ξ_ρ consists of elements of ρ type for R .

Conjecture 1. $\mathcal{F}$ is the free skew-monoidal category on a single generating object via a tensor product causing the faithful functor

$$\text{und} : \mathcal{F} \longrightarrow \Delta_{\text{alg}}$$

$$(u, S) \longmapsto \underline{m}$$

$$\underline{m} = \#S$$

$$\xi \longmapsto \underline{\xi}$$

to be strict monoidal.

Remark The functor $\text{und} : \mathcal{F} \longrightarrow \Delta_{\text{alg}}$ is not conservative. The faithfulness alone does not imply $\mathcal{F}$ monoidal.